



# On 3-choosability of planar graphs with neither adjacent triangles nor 5-, 6- and 9-cycles<sup>☆</sup>

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## ABSTRACT

A graph  $G = (V, E)$  is  $L$ -colorable if for a given list assignment  $L = \{L(v) : v \in V(G)\}$ , there exists a proper coloring  $c$  of  $G$  such that  $c(v) \in L(v)$  for all  $v \in V$ . If  $G$  is  $L$ -colorable for every list assignment  $L$  with  $|L(v)| \geq k$  for all  $v \in V$ , then  $G$  is said to be  $k$ -choosable. In this paper, we prove that every planar graph with neither 5-, 6- and 9-cycles nor adjacent triangles is 3-choosable.

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## 1. Introduction

All graphs considered in this paper are finite, simple planar graphs. A graph  $G$  is *planar* if  $G$  can be drawn on the plane so that its edges meet only at the vertices of the graph. A plane graph is such a particular drawing of a planar graph.

A graph  $G = (V, E)$  is  $L$ -colorable if for a given list assignment  $L = \{L(v) : v \in V(G)\}$ , there exists a proper coloring  $c$  of  $G$  such that  $c(v) \in L(v)$  for all  $v \in V$ . And this coloring is also called an  $L$ -coloring of  $G$ . If  $G$  is  $L$ -colorable for every list assignment  $L$  with  $|L(v)| \geq k$  for all  $v \in V$ , then  $G$  is said to be  $k$ -choosable or  $k$ -list colorable. The *choice number* of  $G$ , denoted by  $\chi_l(G)$  or  $ch(G)$ , is the minimum  $k$  such that  $G$  is  $k$ -choosable.

All 2-choosable graphs were characterized completely in [4]. Thomassen proved that every planar graph is 5-choosable [15]. Examples of planar graphs which are not 4-choosable were given by Voigt [17], and by Mirzakhani [12] independently. Voigt and Wirth [18], and Gutner [7]

independently, presented some planar graphs of girth 4 which are not 3-choosable. Thus, it remains to determine whether a given planar graph of girth 4 is 3- or 4-choosable. In [7], Gutner proved that these problems are NP-complete. Therefore, many authors tried to find sufficient conditions for a planar graph to be 3- or 4-choosable. Alon and Tarsi [1] proved that every planar bipartite graph is 3-choosable. Thomassen [16] proved that every planar graph of girth 5 is 3-choosable. Lam et al. [8,9] proved that plane graphs without  $i$ -cycles are 4-choosable, for  $i = 3, 4, 5$  or 6. Independently, Wang and Lih [20] proved that planar graphs without 5-cycles are 4-choosable. Fijavž et al. [6] proved that planar graphs without 6-cycles are 4-choosable. Farzad [5] showed that a planar graph without 7-cycles is 4-choosable. Xu [23], and Wang and Lih [21] independently, proved that planar graphs without two triangles sharing a common vertex are 4-choosable.

More recently, some new sufficient conditions for a planar graph to be 3-choosable have been given. These include: having no 3-, 5- and 6-cycles [10]; having no 3-, 8- and 9-cycles [25]; having no 3-, 7- and 8-cycles [2] and having no 3-, 6- and 7-cycles [3]. Together with other known parallel results, Wang et al. [22] completed a theorem on 3-choosability of planar graphs: planar graphs without cycles of length 4,  $i, j, 9$  with  $i < j$  and  $i, j \in$

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$\{5, 6, 7, 8\}$  are 3-choosable. Wang [19] proved that every planar graph  $G$  is 3-choosable if it contains no cycle of length at most 10 with a chord.

The distance between two vertices  $x$  and  $y$ , denoted by  $\text{dist}(x, y)$ , is the length of a shortest path connecting them in  $G$ . The distance between two triangles  $T$  and  $T'$  is defined to be the value  $\min\{\text{dist}(x, y) \mid x \in V(T) \text{ and } y \in V(T')\}$ . In [13], Montassier et al. proved that every planar graph either without 4- and 5-cycles and without triangles at distance less than 4, or without 4-, 5- and 6-cycles and without triangles at distance less than 3 is 3-choosable. Zhang and Sun [24] prove that every planar graph with neither 5-, 6- and 7-cycles nor triangles of distance less than 3, or with neither 5-, 6- and 8-cycles nor triangles of distance less than 2 is 3-choosable. In [14] Montassier proposed a conjecture that every planar graph without cycles of length 4, 5, 6, is 3-choosable.

Let  $\mathcal{G}$  denote the set of planar graphs without adjacent triangles and without 5-, 6- and 9-cycles. In this article, we focus on the 3-choosability of graphs in  $\mathcal{G}$ . More precisely, we prove the following result which is an improvement to the result of Zhang and Wu [26]. They prove that every planar graph without any cycle of length 4, 5, 6, or 9 is 3-choosable.

**Theorem 1.** *Every planar graph without adjacent triangles and without 5-, 6- and 9-cycles is 3-choosable.*

We prove Theorem 1 by showing a stronger result as follows. Assuming Theorem 2, one can easily prove Theorem 1.

**Theorem 2.** *Let  $G$  be a planar graph in  $\mathcal{G}$  with  $|L(v)| \geq 3$  for all vertices in  $V(G)$ . If allowing one precolored vertex  $u$  with  $c(u) \in L(u)$ , then this coloring  $c$  can be extended to an  $L$ -list coloring of  $G$ .*

A result of Lovász [11] will be used in the proof of Lemma 3.

**Theorem 3.** (See L. Lovász [11].) *Suppose that a list  $L(x)$  of colours is associated with each vertex  $x$  of a graph  $G$  with  $|L(x)| = d(x)$  for each  $x$  and  $L(v) \neq L(w)$  for some  $v, w$ . Further assume  $G$  to be 2-connected. Then  $G$  admits a proper colouring which uses an element of  $L(x)$  to colour  $x$  for each vertex  $x$ .*

Theorem 3 also means that the only 2-connected graphs which are not  $L$ -list colorable with  $|L(v)| = \Delta(G)$  for all  $v \in V(G)$  are the complete graphs and the odd cycle with identical lists for all vertices.

## 2. Terminology and notation

Let  $G = (V, E, F)$  denote a planar graph, with  $V, E$  and  $F$  being the set of vertices, edges and faces of  $G$ , respectively. We use  $b(f)$  to denote the boundary walk of a face  $f$  and write  $b(f) = [v_1 v_2 v_3 \cdots v_n]$  if  $v_1, v_2, v_3, \dots, v_n$  are the vertices of  $b(f)$  in a cyclic order. A face  $f$  is incident with all vertices and edges on  $b(f)$ . The degree of a face  $f$  of  $G$ , denoted by  $d_G(f)$ , is the number of edges incident

with it, where each cut edge is counted twice. A vertex (face) of degree  $k$  is called a  $k$ -vertex ( $k$ -face). If  $r \leq k$  or  $1 \leq k \leq r$ , then a  $k$ -vertex ( $k$ -face) is called an  $r^+$ - or  $r^-$ -vertex ( $r^+$ - or  $r^-$ -face), respectively. If  $S \subset V(G)$ , then  $G - S$  is the subgraph obtained from  $G$  by deleting the vertices in  $S$  and all the edges incident with some vertices in  $S$ . As usual,  $G[S]$  is the subgraph of  $G$  induced by  $S$ . A chord of a  $k$ -cycle ( $k \geq 4$ ) is an edge joining two non-consecutive vertices on  $C$ .

Two adjacent faces are *normally* adjacent if they have only two vertices in common (clearly, the two common vertices are adjacent), or are *abnormally* adjacent (that is, they have at least three vertices in common). A triangle is synonymous with a 3-cycle.

For  $x \in V(G) \cup F(G)$ , we use  $F_k(x)$  to denote the set of all  $k$ -faces that are incident with or adjacent to  $x$ , and  $V_k(x)$  to denote the set of all  $k$ -vertices that are incident with or adjacent to  $x$ . For convenience, we use  $N(f)$  and  $V(f)$  to denote the set of faces adjacent to the face  $f$  and vertices incident with  $f$  respectively.

## 3. Structures of a minimum counterexample

In this section, we always assume that  $G$  is a counterexample of Theorem 2 with  $|V(G)|$  minimum.  $G$  contains no 5-, 6- and 9-circuits, and adjacent triangles. Clearly,  $|V(G)| \geq 3$  and  $G$  is connected.

**Lemma 1.**  *$G$  does not contain any vertices of degree less than 3.*

**Proof.** Suppose that  $G$  contains a vertex  $u$  of degree less than 3. We choose a vertex  $v$  from  $V(G) \setminus \{u\}$  as the precolored one. By the minimality of  $G$ ,  $G - u$  admits an  $L$ -coloring  $c$ . In  $G$ , we can color  $u$  with a color in  $L(u)$  different from the colors of its neighbors to extend  $c$  to an  $L$ -coloring of  $G$ . This is a contradiction.  $\square$

**Lemma 2.**  *$G$  is 2-connected.*

**Proof.** If  $G$  is not 2-connected, then the block  $B$  containing the precolored vertex is a smaller graph. We color  $B$  first; then we color each block having a vertex in common with  $B$ , etc. This contradicts the assumption that  $G$  is a counterexample.  $\square$

**Lemma 3.** *Any even cycle  $C$  in  $G$  contains at least one  $4^+$ -vertex.*

**Proof.** Let  $V(C) = \{v_1, v_2, \dots, v_{2n}\}$ . Suppose  $d_G(v) = 3$  for all  $v \in C$  by Lemma 1, and  $L$  is a color-list of  $G$  with  $|L(v)| = 3$  for all  $v \in V(G)$ . By assumption, there exists an  $L_0$ -coloring  $\phi_0$  of  $G_0 = G - C$ , where  $L_0$  is the restriction of  $L$  to  $V(G_0)$ . Let  $L' = \{L'(v_i) : 1 \leq i \leq 2n\}$  where  $L'(v_i) = L(v_i) \setminus \{\phi_0(u) : u \in N_G(v_i) \setminus C\}$ .

If  $C$  has no chord: It is clear that  $|L'(v_i)| \geq 2$ . Since every even cycle is 2-choosable, there exists an  $L'$ -coloring  $\phi'$  on  $C$ . An  $L$ -coloring of  $G$  immediately follows by combining  $\phi_0$  and  $\phi'$ . If  $C$  contains chords: It's easy to see in this case  $|L'(v)|$  is equal to the degree of  $v$  in the subgraph of  $G$  induced by  $V(C)$ , and this subgraph is 2-connected and

distinct from  $K_4$ . If there exists a vertex with degree 2 in  $G[C]$ , then we get the proof by Theorem 3. Otherwise, all the vertices in  $V(G[C])$  are of degree 3. We also get an  $L'$ -coloring of  $C$  because  $C$  is an even cycle.  $\square$

**Lemma 4.** *Neither a 3-face nor a 4-face is adjacent to a 4-face.*

**Lemma 5.** *No a 3-face is adjacent to an 8-face.*

**Proof.** Suppose to the contrary that  $G$  has a 3-face  $f$  adjacent to an 8-face  $f_1$ . Let  $f$  be a 3-face with  $b(f) = [v_1 v_2 v_3]$ , and let  $f_1$  be an 8-face with  $b(f_1) = [v_1 v_2 u_1 \times u_2 u_3 u_4 u_5 u_6]$ . If  $f$  is normally adjacent to  $f_1$ , then  $G$  has a 9-cycle. So a 3-face must be abnormally adjacent to the 8-face  $f_1$ . We claim that a 3-face cannot be abnormally adjacent to an 8-face. To show this, we need to prove that  $v_3 \notin V(f_1)$ . Clearly,  $v_3 \neq u_1$ , or we will get that  $\delta(G) \leq 2$ . Next,  $v_3 \neq u_2$  and  $u_3$  since otherwise  $G$  contains a 6-cycle. By similar argument, we also can get that  $v_3 \notin \{u_4, u_5, u_6\}$ .  $\square$

**Lemma 6.** *No a 4-face is adjacent to a 7-face.*

**Proof.** Let  $f_1$  with  $b(f_1) = [v_1 v_2 v_3 v_4]$  be an arbitrary 4-face that is adjacent to a 7-face  $f_2$  with  $b(f_2) = [v_1 v_2 u_1 u_2 u_3 u_4 u_5]$ . If  $f_1$  is normally adjacent to  $f_2$ , then  $G$  has a 9-cycle. So two faces must be abnormally adjacent. Next we will show that  $\{v_3, v_4\} \cap \{u_1, u_2, u_3, u_4, u_5\} = \emptyset$  to complete this proof. First,  $v_3 \neq u_1$ , or there will be a vertex  $v_2$  of degree 2. Second,  $v_3 \neq u_2$ ,  $v_3 \neq u_3$ ,  $v_3 \neq u_4$  and  $v_3 \neq u_5$  since  $G$  has no 5-cycles. So  $v_3 \notin \{u_1, u_2, u_3, u_4, u_5\}$ . Similarly,  $v_4 \neq u_1$ ,  $v_4 \neq u_2$ ,  $v_4 \neq u_3$ ,  $v_4 \neq u_4$  and  $v_4 \neq u_5$  since  $G$  has no 6-cycles. So  $v_4 \notin \{u_1, u_2, u_3, u_4, u_5\}$ .  $\square$

**Lemma 7.** *If  $d(f) = 7$ , then  $|F_3(f)| \leq 1$ .*

**Proof.** Let  $f$  be a 3-face with  $b(f) = [v_1 v_2 v_3]$ , and let  $f_1$  be a 7-face with  $b(f_1) = [v_1 v_2 u_1 u_2 u_3 u_4 u_5]$ . We can prove any 3-face cannot be abnormally adjacent to  $f_1$  or we will get a result which is a contradiction to the fact that  $G$  contains no 5- or 6-cycles as shown above. Assume to the contrary that  $G$  have two 3-face  $f$  and  $f'$  adjacent to the 7-face  $f_1$ . Without loss of generality, suppose that  $b(f) = v_1 v_2 v_3$  and  $b(f') = u_i u_j v_4$ . Because  $G$  contains no adjacent triangles,  $u_i u_j \notin \{v_2 u_1, u_5 v_1\}$ . If  $v_3 \neq v_4$ ,  $G$  contains a 9-cycle, contrary to the assumptions. If  $v_3 = v_4$ , by similar argument, we consider  $u_i u_j = u_3 u_4$  or  $u_i u_j = u_4 u_5$ . In there two cases,  $G$  would contain a 5-cycle, this contradiction completes the proof.  $\square$

#### 4. Proof of Theorem 2

**Proof of Theorem 2.** By contradiction. If the conclusion is not valid, then we choose a counterexample as defined in the last section. We define a weight  $\omega$  on  $V \cup F$  by letting  $\omega(x) = 2d_G(x) - 6$  if  $x \in V$  and  $\omega(x) = d_G(x) - 6$  if  $x \in F$ . By Euler's formula for planar graphs, we have  $\sum_{x \in V \cup F} \omega(x) = -12$ . If we obtain a new nonnegative weight  $\omega^*(x)$  for all  $x \in V \cup F$  by transferring weights from one element to another, then we have

$$-12 = \sum_{x \in V \cup F} \omega(x) = \sum_{x \in V \cup F} \omega^*(x) \geq 0.$$

This contradiction will complete the proof of Theorem 2.

Our transferring rules are as follows.

- (R1) Every  $4^+$ -vertex sends 1 to each incident 3- or 4-face.
- (R2) If  $f_1, f_2, f_3$  are faces incident consecutively with a  $4^+$ -vertex,  $f_1$  is a triangle and  $f_3$  has length at least 7, then  $v$  sends  $\frac{1}{2}$  to  $f_2$ .
- (R3) If a face  $f$  shares an edge  $uv$  with a triangle  $t$ , then  $f$  sends 1 to  $t$  if  $d(u) = d(v) = 3$  and  $\frac{1}{2}$  if  $d(u) = 3$  and  $d(v) > 3$  or vice versa.
- (R4) If a face  $f$  shares an edge  $uv$  with a 4-face  $t$  and  $d(u) = d(v) = 3$ , then  $f$  sends  $\frac{1}{2}$  to  $t$ .

Now, we proceed to calculating  $w^*(x)$  for  $x \in V(G) \cup F(G)$ .

Let  $v$  be a  $k$ -vertex of  $G$ . Then  $k \geq 3$  by Lemma 1.

If  $k = 3$ , it is clear that  $w^*(v) = w(v) = 2d(v) - 6 = 0$ .

If  $k = 4$ , then  $w(v) = 2$ . We can see that  $|F_3(v)| + |F_4(v)| \leq 2$  by Lemma 4. If  $|F_3(v)| + |F_4(v)| \leq 1$ , then  $w^*(v) \geq w(v) - 1 - 2 \times \frac{1}{2} = 0$  by (R1) and (R2). If  $|F_3(v)| + |F_4(v)| = 2$ , we get that  $w^*(v) = w(v) - 2 \times 1 = 0$  by (R1) and (R2).

If  $k \geq 5$ , note that any vertex  $v$  sends at most 2 to each three consecutive faces incident with it, so  $w^*(v) \geq w(v) - 2 \times \lfloor \frac{d(v)}{3} \rfloor = \frac{4}{3}d - 6 > 0$ .

Let  $f$  be an  $h$ -face of  $G$ .

If  $h = 3$  or  $h = 4$ , it is easy to see the charge of 3- and 4-faces is nonnegative by (R1), (R3) and (R4).

If  $h = 7$ , then  $w(f) = 1$ . By Lemmas 6 and 7, a 7-face  $f$  only sends at most 1 to the adjacent triangle by (R3). This fact means  $w^*(f) \geq w(f) - 1 = 0$ .

If  $h = 8$ , then  $w(f) = 2$ . By Lemma 5,  $f$  is not adjacent to any 3-faces. So an 8-face  $f$  sends nothing to 3-faces by (R3). Moreover, an 8-face  $f$  sends totally at most  $4 \times \frac{1}{2}$  to adjacent 4-faces by (R4) and Lemma 4, thus  $w^*(f) \geq w(f) - 4 \times \frac{1}{2} = 0$ .

If  $h = 10$ , then  $w(f) = 4$ . The only way a 10-face  $f$  could have a negative charge would be if all its vertices had degree 3, but this case is excluded by Lemma 3, or exactly one has degree at least 4, and  $f$  would be adjacent to 5 triangles. In this case  $f$  receives  $1/2$  by (R2), making its charge 0.

If  $h = 11$ , then  $w(f) = 5$ .  $f$  is adjacent to at most 5 triangles. We have  $w^*(f) = 11 - 6 - 1 \times 5 = 0$  by (R3).

If  $h \geq 12$ , then  $w(f) \geq 6$ . Observe that by Lemma 4 and the rules (R3) and (R4), we consider the case that the number of adjacent triangles is as more as possible, thus  $w^*(f) \geq w(f) - 1 \times \lfloor d/2 \rfloor \geq 0$ .  $\square$

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